

Tatenda A. Tobaiwa

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Software Engineer • MSc Candidate in Computer Science • Computational Geometer

ABOUT ME

I am a software engineer, specifically C programming and computational geometer who finds genuine joy at the intersection of rigorous mathematics and real-world impact. My doctoral-track research asks a deceptively simple question: *how do we reconstruct a complete, near-real-time view of the Earth from a swarm of small satellites in under ten minutes?* The answer involves orbital mechanics, hexagonal spatial indexing, and deterministic set-cover optimisation — and that mix of disciplines is exactly where I feel most alive professionally.

Beyond research, I have built production web systems, coached undergraduate students through their first pointers and recursive algorithms, and won two national hackathon championships. I am driven by a conviction that advanced technology, built thoughtfully and deployed responsibly, can make a tangible difference in communities across southern Africa — from flood-monitoring in the Okavango Delta to drought response in the Kgalagadi.

EDUCATION

Master of Science in Computer Science

Botswana International University of Science & Technology (BIUST)

Palapye, Botswana

Aug 2025 – Present

- Supervised by Prof. Irina Zlotnikova and Dr. Dimane Mpoeleng.
- Research focus: *Algorithmic Reconstruction of Near-Real-Time Mosaics from Distributed LEO Satellite Imagery* — combining Computational Geometry (H3 DGGS), astrodynamics (Orekit/SGP4), and combinatorial optimisation.
- Presented original research at the African Astronomical Society (AfAS) Conference 2026.

Bachelor of Science in Computer Software Engineering

Botswana International University of Science & Technology (BIUST)

Palapye, Botswana

Aug 2021 – May 2025

- Graduated with a strong foundation spanning algorithms, systems programming, and machine learning — reflected in an independent deep-learning pipeline that achieved **99.57%** classification accuracy on a clinical cancer dataset.

AWARDS & RECOGNITION

1st Place — Huawei Innovation Challenge

National Champion & Regional Finalist (2025/2026). Architected a highly scalable technical solution representing Botswana at the regional stage.

1st Place — Orange Digital Hackathon

National Champion. Delivered an innovative full-stack prototype under strict time pressure, demonstrating elite rapid-development and problem-solving skills.

Conference Presenter — AfAS 2026

Presented MSc research at the African Astronomical Society Conference, contributing to the growing body of African-led satellite remote-sensing science.

PROFESSIONAL EXPERIENCE

Teaching Assistant — Computer Science

Botswana International University of Science & Technology

Palapye, Botswana

Aug 2025 – Present

- Data Structures & Algorithms:** Guide students from first principles through algorithmic complexity ($\mathcal{O}$ -notation, amortised analysis), graph theory, and optimisation heuristics — bridging the gap between mathematical abstraction and efficient, maintainable code.
- C Programming:** Mentor students in low-level systems programming: memory management, pointer arithmetic, and disciplined debugging — skills that remain foundational regardless of the technology stack.
- Software Testing & Analysis:** Teach the full testing lifecycle, including automated test-suite development with Selenium (Java and Python bindings) and practical test-design strategies for industry-grade codebases.

Software Engineering Intern

Spectrum Analytics

Gaborone, Botswana

May 2024 – Jul 2024

- Transitioned from academic development into a high-pressure enterprise setting, rapidly adapting to production-grade engineering standards and team workflows.
- Contributed to robust ETL data pipelines and client-facing web infrastructure, collaborating directly with senior data architects on cross-functional delivery teams.
- Gained practical exposure to cloud infrastructure patterns and enterprise data-fabric design — experience that now directly

informs the scalability considerations of my MSc system architecture.

Full-Stack Web Developer
Freelance / Self-Employed

Francistown, Botswana
Nov 2024 – Present

- Design, build, and deploy responsive, accessible web applications for diverse clients, leveraging modern JavaScript frameworks with a focus on user experience, performance, and maintainable backend architectures.
- Manage the full project lifecycle independently — from client briefing and technical specification through to deployment and ongoing support.

KEY PROJECTS & VOLUNTEER WORK

Algorithmic LEO Mosaic Reconstruction (MSc Thesis)
Python Orekit H3-py SGP4 NumPy

Lead Researcher
2025 – Present

- Designing a **deterministic set-cover optimisation algorithm** to reconstruct country-scale, near-real-time satellite mosaics — targeting a latency reduction from the current industry baseline of 72 hours to **under 10 minutes**.
- Propagates LEO orbital elements via SGP4 within the Orekit astrodynamics library and discretises Earth’s surface using Uber’s H3 hexagonal DGGS, enabling $\mathcal{O}(n \log n)$ spatial set-cover formulations with provable coverage guarantees.
- Introduces *Fail-Signal semantics* to handle cloud-obscured data gaps deterministically — turning an ambiguous missing-data problem into a well-defined algorithmic signal that downstream consumers can act upon.
- Applied context: flood-inundation monitoring in the Okavango Delta and fire-scar mapping in the Waterberg Biosphere Reserve, Botswana.

Lung Cancer Classification via Deep Learning
Python TensorFlow Keras Grad-CAM LIME

Lead Developer
Apr 2025 – May 2025

- Engineered a two-phase transfer-learning pipeline ensembling four pre-trained CNNs (DenseNet121, InceptionV3, ResNet50V2, MobileNetV2), achieving **99.57% accuracy** and **100% recall** on malignant cases.
- Integrated Explainable AI (XAI) techniques — Grad-CAM saliency maps and LIME perturbation analysis — to surface clinically interpretable visual evidence for each prediction, addressing the trust and transparency requirements of medical AI deployment.

Ablehearts Foundation Website
HTML/CSS JavaScript Responsive Design UI/UX

Volunteer Full-Stack Developer
Sep 2024 – Present

- Designed and launched the official website for a non-profit organisation supporting underprivileged children in Botswana — translating a social mission into a functional, emotionally resonant digital platform.
- Built to drive donation campaigns and raise community awareness, with accessibility and mobile-first design as first-class requirements.

TECHNICAL SKILLS

Domain	Details
Languages & Frameworks	Python, C, Java, JavaScript, Selenium (Java/Python), TensorFlow, Keras
Space & Geometry	Astrodynamics (Orekit, SGP4), Computational Geometry (H3 DGGS), Orbital Propagation
Machine Learning / AI	CNNs, Transfer Learning, Ensemble Methods, Explainable AI (Grad-CAM, LIME)
Software Engineering	Data Structures & Algorithms, Test Automation (Selenium), Agile Workflows, Git
Tools & Platforms	Google Colab, CelesTrak, H3-py, NumPy, Pandas, Linux

CERTIFICATIONS

- Leading with Character
- Career Essentials in Software Development
- Introduction to Generative AI
- Introduction to Responsible AI
- Email Marketing Professional